Question			Marking details]	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	*(a)*		Work / charge **or** energy / charge **or** energy / coulomb (1) Don't accept joules / coulomb Context i.e. energy transferred [or converted] from electrical [potential] / work done………. [between the points] not a freestanding mark (1)	2			2		
	(b)	(i)	1.62 [V] **or** emf	1			1		
		(ii)	$I = \frac{1.38}{1.50}$ [=0.92 A] **or** by implication if answer correct to 2 sf (1) $V_r = 1.62\text{ V} - 1.38\text{ V} = [0.24\text{ V}]$ **or** $R + r = \frac{1.62\text{ V}}{0.92\text{ A}} = [1.76\ \Omega]$(1) $r = 0.26\ [\Omega]$ (1)		3		3	3	
		(iii)	$t = \frac{750\text{ J}}{EI}$ *(*1) or by implication **ecf** $t = 500\text{ s}\ \ [503\text{ s}]$ (1) [590 s indicates wrong pd used so award 1 mark only]		2		2	2	
		(iv)	Total circuit resistance = 0.26 Ω **ecf** + 0.75 Ω or by implication (1) $V_R = 1.20$ [V] (1) or 0.3 Ω + 0.75 Ω gives $V_R = 1.16$ [V]		2		2	2	
	(c)	(i)	$I = \frac{6.0\text{ V}}{200\ \Omega + 850\ \Omega)}$ [= 5.71 mA] **or** $V_{200} = \frac{200}{200+850}\ 6.0\text{ V}$ (1) $V_{200} = 1.14$ [V] (1)V_{LDR} = 4.86 [V] but V_{200} not given award 1 mark		2		2	2	
		(ii)	**No credit** for 'Resistance decreases' Current increases **or** *total* circuit resistance decreases **or** any relevant resistance ratio (e.g. $\frac{V_{200}}{V_{\text{LDR}}}$) stated to increase or decrease as appropriate to the ratio (1) V_{200} goes up, but award mark only if argument presented so that this conclusion follows (1) **no ecf** from (c)(i)		2		2		
			Question 5 total	**3**	**11**	**0**	**14**	**9**	**0**